📝 GLA UNIVERSITY // DAY 1 IN-CLASS CONCEPT CHECKPOINT (STUDENT QUESTION PAPER)

High Performance Computing & Heterogeneous AI Systems (`GLA-HPC-BCSG701`)

Date: Day 1 Masterclass | **Duration:** Active Lecture Checkpoints | **Total Marks:** 25 Marks

Instructions for Students:

1. Select the single best answer (a, b, c, or d) for each question as presented during the lecture.
2. Note your reasoning before the instructor reveals the architectural mechanism.

Question 01: Why do large AI models (e.g., 70-Billion parameter LLMs) fail to train efficiently on high-end consumer laptops despite having fast CPUs?

- **a)** Laptops lack sufficient DDR memory channels and bandwidth (choked by the Memory Wall)
- **b)** Laptops do not support floating-point arithmetic
- **c)** Windows OS permanently limits memory allocation to 4 GB
- **d)** Laptop CPUs cannot execute C++ code

Question 02: Under Flynn's classical computer architecture taxonomy, which architectural model best describes modern GPUs and CPU vector execution units (AVX-512)?

- **a)** SISD (Single Instruction, Single Data)
- **b)** SIMD (Single Instruction, Multiple Data)
- **c)** MISD (Multiple Instruction, Single Data)
- **d)** MIMD (Multiple Instruction, Multiple Data)

Question 03: An HPC simulation contains a sequential fraction of 10% that cannot be parallelized. According to Amdahl's Law, what is the theoretical maximum speedup limit, even with an infinite number of processor cores?

- **a)** 100x speedup
- **b)** 10x speedup
- **c)** 50x speedup
- **d)** Infinite speedup

Question 04: In an enterprise heterogeneous supercomputing cluster, what is the primary architectural responsibility of Intel Xeon Scalable processors?

- **a)** High-latency batch graphics rendering only
- **b)** Orchestrating workflows, managing OS scheduling, preprocessing tensors, and handling irregular sequential logic
- **c)** Replacing all network switches in the datacenter

- **d)** Acting exclusively as passive memory controllers

Question 05: What specific hardware capability is introduced on 4th Gen Intel Xeon Scalable processors (Sapphire Rapids) via Intel AMX (Advanced Matrix Extensions)?

- **a)** Integrated liquid nitrogen cooling controllers
- **b)** Dedicated 2D tile registers (TMM0-TMM7) and matrix multiplication units (TMUL) accelerating deep learning on the CPU die
- **c)** Hardware emulation of 8-bit retro video game graphics
- **d)** Direct connection to SATA-1 mechanical hard drives

Question 06: Which computational workload is the Intel Data Center GPU Max Series (formerly code-named Ponte Vecchio) primarily architected to accelerate?

- **a)** Massive parallel floating-point vector/matrix simulations and high-density FP64/FP32 scientific computing
- **b)** Single-threaded shell script execution
- **c)** Low-speed serial ASCII text parsing
- **d)** Managing Linux user account passwords

Question 07: How does the Habana Gaudi 2 AI accelerator achieve scale-out multi-node cluster training without requiring external discrete InfiniBand PCIe host channel adapters (HCAs)?

- **a)** By transmitting data across standard USB 2.0 cables
- **b)** By integrating 24x 100GbE RoCE (RDMA over Converged Ethernet) ports directly onto each Gaudi silicon die
- **c)** By compressing all neural weights into local CPU L1 cache
- **d)** By running software emulated WiFi bridges

Question 08: Which fundamental design philosophy of the Linux operating system enables hardware devices, running processes, and network sockets to be manipulated using standard file commands (cat, read, write)?

- **a)** Everything is an executable script
- **b)** Everything is a file (Virtual File System abstraction)
- **c)** Everything is a Windows registry entry
- **d)** Everything is a shared cloud database

Question 09: An HPC cluster administrator runs 'chmod 750 run_simulation.sh'. What exact permissions are granted to Owner, Group, and Others?

- **a)** Owner: rwx, Group: r-x, Others: ---
- **b)** Owner: rw-, Group: r--, Others: r--
- **c)** Owner: rwx, Group: rwx, Others: rwx
- **d)** Owner: r-x, Group: ---, Others: ---

Question 10: If an HPC user's shell profile sets 'umask 027', what default octal permissions will be assigned to newly created directories and text files respectively?

- **a)** Directory: 750 (rwxr-x---), File: 640 (rw-r-----)
- **b)** Directory: 777 (rwxrwxrwx), File: 666 (rw-rw-rw-)
- **c)** Directory: 755 (rwxr-xr-x), File: 644 (rw-r--r--)
- **d)** Directory: 700 (rwx------), File: 600 (rw-------)

Question 11: In Linux cluster navigation, what does the command 'cd ../../data' instruct the shell to do?

- **a)** Search for the word 'data' across all files in the system
- **b)** Move up two directory levels from the current location, then enter the 'data' directory
- **c)** Delete two parent folders and format the disk
- **d)** Open a graphical file manager window

Question 12: Why is '#!/usr/bin/env bash' widely considered a best practice over hardcoded '#!/bin/bash' on enterprise HPC clusters?

- **a)** It compiles the bash script into machine code before running
- **b)** It dynamically searches the user's $PATH to locate the bash binary across diverse Linux distributions
- **c)** It forces bash to run with superuser (root) privileges
- **d)** It automatically translates bash code into Python

Question 13: What is the critical defensive engineering purpose of placing 'set -euo pipefail' at the beginning of an automated cluster orchestration script?

- **a)** It increases CPU clock frequency by 15%
- **b)** It causes the script to abort immediately if any command errors (-e), any unset variable is used (-u), or any pipe step fails (pipefail)
- **c)** It ignores all syntax errors and suppresses terminal logging
- **d)** It encrypts the script using AES-256

Question 14: Why will the shell command 'BATCH_SIZE = 64' produce a 'command not found' error in Bash?

- **a)** Bash does not support numbers larger than 32
- **b)** Bash treats spaces around '=' as argument delimiters, interpreting 'BATCH_SIZE' as a command name
- **c)** Variable names in bash must start with a dollar sign ($)
- **d)** Variables can only be declared inside compiled C++ code

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Question 15: In the command './hpc_node_audit.sh > audit.log 2>&1', what is the exact effect of '2>&1'?

- **a)** It multiplies the output volume by 2
- **b)** It redirects standard error (file descriptor 2) to the same destination as standard output (file descriptor 1)
- **c)** It runs the script twice on core 1
- **d)** It delays script execution by 2 seconds

Question 16: When generating cluster access credentials via 'ssh-keygen -t ed25519', why is the Ed25519 algorithm preferred over standard RSA 2048/4096?

- **a)** Ed25519 uses shorter 256-bit elliptic-curve keys, achieves faster handshakes, and provides superior resistance to side-channel attacks
- **b)** RSA is incompatible with Linux operating systems
- **c)** Ed25519 does not require a private key
- **d)** Ed25519 allows passwords to be transmitted in plain text

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Question 17: To enable passwordless automated job submission from a local machine to a remote cluster node, into which remote file must your client's public key (~/.ssh/id_ed25519.pub) be appended?

- **a)** /etc/passwd
- **b)** ~/.ssh/authorized_keys
- **c)** ~/.ssh/known_hosts
- **d)** /tmp/public_keys.txt

Question 18: A student attempts to connect to an HPC node and receives: 'Permissions 0644 for id_ed25519 are too open. It is required that your private key files are NOT accessible by others.' What command resolves this?

- **a)** chmod 777 ~/.ssh/id_ed25519
- **b)** chmod 600 ~/.ssh/id_ed25519
- **c)** rm -rf ~/.ssh
- **d)** chmod 000 ~/.ssh/id_ed25519

Question 19: What is the primary industry objective of the Intel oneAPI software ecosystem?

- **a)** To replace the Linux kernel with a custom proprietary operating system
- **b)** To provide an open, standards-based unified programming model across heterogeneous architectures (CPUs, GPUs, FPGAs) without vendor lock-in
- **c)** To convert Python scripts directly into hardware circuits
- **d)** To eliminate the need for computer RAM

Question 20: What is SYCL (pronounced 'sickle') in the context of high performance computing with Intel oneAPI?

- **a)** A high-level Khronos standard C++ programming model for single-source heterogeneous programming
- **b)** A relational database query engine
- **c)** An alternative to the Linux Bash shell
- **d)** A network cabling standard for InfiniBand

Question 21: In SYCL/DPC++, what is the principal architectural advantage of allocating memory via 'sycl::malloc_shared' (Unified Shared Memory)?

- **a)** It doubles the available storage capacity of hard disks

- **b)** It provides a single virtual pointer accessible by both CPU (Host) and GPU/Accelerator (Device) with automatic page migration
- **c)** It permanently disables CPU cache coherence
- **d)** It translates floating point numbers into integers

Question 22: Which shell command initializes the Intel oneAPI development environment, populating $PATH, $LD_LIBRARY_PATH, and compiler binaries (icpx, ifx)?

- **a)** source /opt/intel/oneapi/setvars.sh
- **b)** python3 /opt/intel/install.py
- **c)** systemctl restart oneapi
- **d)** apt install oneapi-now

Question 23: What is the defining distinction between the Intel oneAPI Base Toolkit and the Intel HPC Toolkit?

- **a)** Base Toolkit is for single-node development (SYCL, MKL, DAL); HPC Toolkit adds multi-node cluster tools (Intel MPI, Fortran ifx, Inspector, ITAC)
- **b)** Base Toolkit is paid commercial software; HPC Toolkit is open-source
- **c)** Base Toolkit is for Python; HPC Toolkit is for Java
- **d)** There is no difference; they are exact duplicates

Question 24: Which Linux terminal utility provides immediate introspection of CPU architecture, hardware thread count, cache hierarchy (L1/L2/L3), and NUMA topology?

- **a)** lscpu
- **b)** ping
- **c)** chmod
- **d)** grep

Question 25: When compiling an HPC benchmark with the Intel oneAPI C++ Compiler (icpx), which optimization flag instructs the compiler to generate instructions matching the host processor's highest SIMD capability (e.g., AVX-512)?

- **a)** -xHost
- **b)** -O0
- **c)** -w
- **d)** -g
